

1. Administrative & Study Identification

Full Study Title A Study Evaluating the Efficacy and Safety of Crovalimab Versus Eculizumab in Participants With Paroxysmal Nocturnal Hemoglobinuria (PNH) Not Previously Treated With Complement Inhibitors **Short Title/Acronym** COMMODE 2 **Protocol Number** B042162 **NCT Number** NCT04434092 **Trial Status** Active, Not Recruiting **Sponsor Name** Roche (Hoffmann-La Roche) **Investigational Product** Crovalimab (PiaSky®) / Active Comparator: Eculizumab (Soliris®) **Phase of Development** Phase III **Medical Monitor** Roche Clinical Operations Lead / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the non-inferiority of crovalimab compared with eculizumab in achieving hemolysis control and transfusion avoidance in patients with paroxysmal nocturnal hemoglobinuria (PNH) who have not been previously treated with C5 complement inhibitors. **Secondary Objectives:** To evaluate the proportion of patients with breakthrough hemolysis, stabilized hemoglobin, and clinically meaningful changes in the FACIT-Fatigue score, alongside long-term safety.

2.2 Background & Rationale The current standard of care for PNH involves C5 inhibition (e.g., eculizumab), which requires frequent, time-consuming intravenous infusions, adding a significant burden to patients. Crovalimab is a novel, subcutaneously administered C5 complement inhibitor that enables rapid and sustained C5 inhibition with a low-volume, weight-based self-administration every 4 weeks. This study was conducted to demonstrate that crovalimab is non-inferior to eculizumab, offering patients greater freedom, flexibility, and a reduced treatment burden without compromising efficacy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Eculizumab) **Blinding:** Open-label **Randomization:** 2:1 (Crovalimab to Eculizumab)

3.2 Planned Enrollment & Duration Target \$N\$: 210 patients targeted **Number of Sites:** Global, Multicenter **Estimated Study Start:** 08-Oct-2020 **Estimated Primary Completion:** 16-Nov-2022 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Documented diagnosis of PNH, confirmed by high sensitivity flow cytometry. 2. LDH level $\geq$ 2x the upper limit of normal (ULN) at screening (as per local assessment). 3. Body weight $\geq$ 40 kg at screening (pediatric participants with body weight < 40 kg). 4. Vaccination against *Neisseria meningitidis* serotypes A, C, W, and Y $\leq$ 3 years prior to initiation of study treatment; or, if not previously done, vaccination administered no later than one week after the first drug administration. 5. Willingness and ability to comply with all study visits and procedures. 6. Women of childbearing potential: agreement to remain abstinent or use contraception during the treatment period and for 10.5 months after the final dose of crovalimab or for 3 months after the final dose of eculizumab (or longer if required by the local product label).

4.2 Exclusion Criteria 1. Current or previous treatment with a complement inhibitor. 2. History of allogeneic bone marrow transplantation. 3. History of *Neisseria meningitidis* infection within 6 months prior to screening and up to first study drug administration. 4. History of myelodysplastic syndrome with Revised International Prognostic Scoring System (IPSS-R) prognostic risk categories of intermediate, high and very high. 5. Pregnant or breastfeeding, or intending to become pregnant during the study, within 10.5 months after the final dose of crovalimab, or 3 months after the final dose of eculizumab (or longer if required by the local product label). 6. Participation in another interventional treatment study with an investigational agent or use of any experimental therapy within 28 days of screening or within 5 half-lives of that investigational product, whichever is greater. 7. Concurrent disease, treatment, procedure or surgery, or abnormality in clinical laboratory tests that could interfere with the conduct of the study, may pose any additional risk for the participant, or would, in the opinion of the Investigator, preclude the participant's safe participation in and completion of the study. 8. Splenectomy < 6 months before screening. 9. Positive for Active Hepatitis B and C infection (HBV/HCV). 10. History of or ongoing cryoglobulinemia at screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Crovalimab:** Weight-based tiered regimen. Intravenous (IV) loading dose on Day 1, followed by weekly subcutaneous (SC) doses for 4 weeks. Maintenance begins at Week 5 with SC injections every 4 weeks (Q4W). * **Eculizumab (Soliris®):** IV infusion weekly for the first 4 weeks, followed by a maintenance IV infusion of 900 mg every 2 weeks. **Route of Administration:** Subcutaneous (SC) for crovalimab maintenance; Intravenous (IV) for eculizumab. **Duration of**

Treatment: 24-week primary treatment period, followed by an optional extension period of up to 5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for PNH, including red blood cell transfusions if required. **Prohibited:** Other complement inhibitors or investigational therapies during the primary 24-week treatment period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, PNH Diagnosis Confirmation, LDH Labs, Meningococcal Vaccination Verification
Baseline	Day 1	Randomization, First IV Loading Dose (Crovalimab or Eculizumab), Baseline FACIT-F
Interim	Weeks 1-4	Weekly loading doses (SC for Crovalimab / IV for Eculizumab), Safety monitoring
Maintenance	Weeks 5-24	Q4W SC injections (Crovalimab) or Q2W IV infusions (Eculizumab), LDH Assessment, FACIT-F
End of Study	Week 25	Final primary endpoint evaluation (Hemolysis control & Transfusion avoidance), Option to enter Extension Period

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- * **Percentage of Participants With Transfusion Avoidance (TA):** TA is defined as participants who are packed red blood cell (pRBC) transfusion-free and do not require transfusion per protocol-specified guidelines from baseline to week 25. (95% CI calculated using Wilson's method with continuity correction).
- * **Percentage of Participants With Hemolysis Control Measured by LDH:** Evaluated from week 5 to week 25, defined as $LDH \leq 1.5 \times ULN$. (Generalized Estimating Equation (GEE) used to estimate the population-average percentage, taking into account intra-patient and inter-patient correlation).

7.2 Secondary Endpoints

- * Percentage of participants with breakthrough hemolysis (BTH).
- * Percentage of participants with stabilization of hemoglobin (avoidance of a ≥ 2 g/dL decrease in hemoglobin level from baseline in the absence of transfusion).
- * Change from baseline in fatigue assessed by Functional Assessment of Chronic Illness Therapy-Fatigue (FACIT-F) Scale.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of untoward medical occurrences, with specific surveillance for injection-site reactions, transient immune complex reactions (TICRs), and severe infections (e.g., *Neisseria meningitidis*). **Serious Adverse Events (SAEs):** Standard reporting timelines applied for SAEs and

fatal AEs. **Data Monitoring Committee (DMC):** Independent safety oversight committee managed by the sponsor to evaluate the risk-benefit profile.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=210$ appropriately powered to test the non-inferiority margin of crovalimab versus eculizumab via odds ratios and weighted differences for co-primary endpoints. **Handling of Missing Data:** Participants who withdrew before Week 25 were deemed to have had a transfusion and not have had hemoglobin stabilization.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to assure data integrity and patient safety. **Audit Trail:** Data collected via eCRF locking mechanisms with data clarification forms as per standard quality audit procedures.

11. Study Identifiers & Governance

Posting ID: 687894870374806523 **Primary ID:** B042162 **Registry Number:** NCT04434092 **Sponsor/Lead Organization:** Roche **Study Title:** COMMODORE 2 **Official Regulatory Title:** A Phase III, Randomized, Open-label, Active-controlled, Multicenter Study Evaluating the Efficacy and Safety of Crovalimab Versus Eculizumab in Patients With Paroxysmal Nocturnal Hemoglobinuria (PNH) Not Previously Treated With Complement Inhibitors

12. Clinical Framework (PNH Specific)

Primary Condition: Paroxysmal Nocturnal Hemoglobinuria (PNH) **Diagnosis Threshold:** LDH $\geq 2.0 \times \text{ULN}$ and high-sensitivity flow cytometry positive **Medical Methodology:** C5 Complement Inhibition (Crovalimab vs. Eculizumab) **Optimization Target:** Hemolysis control (LDH $\leq 1.5 \times \text{ULN}$) and Transfusion Avoidance

13. Study Procedures & Methodology

Management Pathway: Active complement inhibition treatment pathway. **Education Protocol (HCP):** Training on weight-based tiered dosing and identification of breakthrough hemolysis or

TICRs. **Education Protocol (Patient):** Onsite training for crovalimab SC self-administration (or caregiver administration) starting from Week 9. **Quality Audit Mechanism:** Centralized evaluation of LDH and adherence to the self-administration protocol.

14. Observation & Follow-Up Schedule

Total Duration: 24-week primary treatment period + optional extension up to 5 years. **Onsite Visit Cadence:** Weeks 1, 2, 3, 4, and regular intervals corresponding to the Q4W (Crovalimab) or Q2W (Eculizumab) dosing schedule. **Remote Monitoring Frequency:** Follow-up contacts for safety events and medication compliance. **Checklist Review Interval:** Regular administration of the 13-item FACIT-F questionnaire to track fatigue severity.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	____	[DD-MMM-YYYY]				